

Challenge (Habanero)

- Q1.** If $f(x) = x^2$ and $g(x) = 2x$, what is $(f + g)(x)$?
(A) $x^2 + 2x$
(B) $x^2 - 2x$
(C) $2x^2$
(D) $x^2 + x$
(E) $2x^2 + x$
- Q2.** A hiker's fuel consumption is modeled by $f(x) = 2x$ gallons per mile, and her backpack's weight is $g(x) = x^2$ pounds. What is $(f \cdot g)(x)$?
(A) $2x^3$
(B) $x^2 + 2x$
(C) $2x^2$
(D) $2x^2 + x$
(E) x^3
- Q3.** Given $f(x) = x^2 - 1$ and $g(x) = x + 1$, what is the domain of $(f/g)(x)$?
(A) $(-\infty, -1) \cup (-1, \infty)$
(B) $(-\infty, 1) \cup (1, \infty)$
(C) $(-\infty, -1]$
(D) $[-1, \infty)$
(E) $(-1, 1)$
- Q4.** What is $(f - g)(x)$ if $f(x) = 3x$ and $g(x) = x^2 - 2x$?
(A) $x^2 - 5x$
(B) $-x^2 + 5x$
(C) $3x - x^2$
(D) $3x + x^2$
(E) $-3x + x^2$
- Q5.** A park's animal population is given by $f(x) = 100 + 2x$ and the number of plants is $g(x) = 50x - x^2$. What is $(f + g)(x)$?
(A) $50x + 100 - x^2$
(B) $100 + 2x + 50x - x^2$
(C) $50x - 100 - x^2$
(D) $2x + x^2$
(E) $100 + x^2$
- Q6.** What is the simplified form of $(f \cdot g)(x)$ if $f(x) = x + 2$ and $g(x) = x^2 - 4$?
(A) $x^3 - 4x + 2x^2 - 8$
(B) $x^3 + 2x^2 - 4x - 8$
(C) $x^3 - 2x^2 - 4x + 8$
(D) $x^3 + 2x^2 - 4x - 2$
(E) $x^3 - 4x - 8$
- Q7.** Given $f(x) = 2x - 1$ and $g(x) = x^2 + x - 1$, what is $(f - g)(x)$?
(A) $-x^2 + x + 1$
(B) $-x^2 - x + 2$
(C) $2x - 1 - x^2 - x + 1$
(D) $-x^2 - x - 1$
(E) $x^2 + 2x - 1$
- Q8.** If $f(x) = x^2$ and $g(x) = x - 2$, what is $(f/g)(x)$?
(A) $\frac{x^2}{x-2}$
(B) $x^2 + 2$
(C) $x + 2$
(D) $x^2 - 4$
(E) $x - 2$

Open Ended Questions (FRQ)

- Q9.** Suppose the population of a certain animal species in a park is modeled by $f(x) = 200 + 5x$ and the park's area is $g(x) = x^2 - 10x + 25$. Determine the expression for $(f \cdot g)(x)$ and simplify. Show all your steps and calculations. You may use the back of the page if necessary.
- Q10.** For the functions $f(x) = 3x - 2$ and $g(x) = x^2 + x - 1$, find the domain and the simplified expression of $(f/g)(x)$. Explain your reasoning and show your work.

Chef's Tips

- Provide clear instructions for each type of question
- Ensure students have access to necessary mathematical tools and formulas
- Allow students to ask for clarification if they are unsure about any part of the test